[Project Title: Insert Full Title Here]

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A Research Report Submitted for the
Regeneron Science Talent Search

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Abstract

[Insert Abstract Here — 250 words maximum. Provide a concise summary of the research question, core hypothesis or objective, computational or experimental methodology, quantitative results, and broader implications of the findings.]

1 Background of Portfolio Construction

1.1 Principles of Portfolio Construction

Modern portfolio construction involves optimizing a weight vector $w \in \mathbb{R}^n$ over n assets. To assess the optimality of a portfolio, an objective function is used. An example of a classic objective function is Markowitz's Mean-Variance framework, which maximizes a risk-adjusted objective:

$$\min_w(\mu^T w - \lambda w^T \Sigma w), \tag{1}$$

where μ is an expected returns vector, λ is an exogenous risk-aversion parameter, and Σ is a covariance matrix that defines how assets move together.

Objective functions can be comprised of other forms, such as returning an L_2 norm of the difference between a calculated target weight vector $\tilde{w}$ and w.

$$f(w) = \min_w(||\tilde{w} - w||_2^2) \tag{2}$$

This objective function will be elaborated upon in section ADDREF.

Constraints encode what is permissible when calculating optimal potfolio weights. Typical constraints include individual weight and total potfolio turnovers, position limits, etc. Implementing such constraints often make portfolio optimization non-trivial, and can present difficulties with convexity and proper execution.

Penalties occupy a middle-ground between an objective function and constraints; instead of placing a hard boundary on portfolios, they discourage going past a certain threshold continuously. In the case of 2, penalties come in the form of added terms.

Once the objective function, constraints, and penalties are defined, the choice of solving method will be contingent on the problem's structure. Objective 1 is convex if Σ is positive semi-definite and if its constraints maintain a convex feasible set. Objective 2 acts similarly, placing both objectives within the reach of Disciplined Convex Programming (DCP). The convexity makes any local optimum a global minimum or maximum, and the Karush-Kuhn Tucker (KKT) conditions provide a complete characterizastion of optimality which we will elaborate upon later in section ADDREF.

The widespread efficiency and availability of DCP solvers means convex problems can be optimized notably quickly.

1.2 Research gap related to constraints and penalties

2 Problem Setup

2.1 L_2 Target-Optimize Vectors

Let $\tilde{w}$ be a weight vector with the same size as w. It is a signal-driven constant vector indicating ideal target weights to consider in constructing a new portfolio. $\tilde{w}$ is normalized such that it possesses a consistent scale and is comparable across adjustments. In long-short portfolios, this typically takes the form of a gross

exposure normalization:

$$\mathbf{1}^T|\tilde{w}| = \Omega,$$
$$\tilde{w} = \frac{\tilde{w}}{\Omega}$$

Normalizing fixes scales and units in which the objective function and subsequent constraints are expressed, and avoids degenerate solutions where magnitudes are dilated rather than truly altering relative composition of the portfolio.

Gross exposure constraints will also be placed upon w, the new portfolio being optimized. With exact gross exposure constraints that apply to w and $\tilde{w}$, this ensures equal scale. This gross exposure constraint must be an equality to be effective, and is elaborated upon in ADDREF.

The primary objective function $f(w)$ we will use for the remainder of this paper will be

$$f(w) = ||\tilde{w} - w||_2^2 + (\dots), \tag{3}$$

where $(\dots)$ indicate additional penalties added to the objective (to be added in sections ADDREF) $f(w)$ is currently quadratic and convex, meaning it can be optimized through Quadratic Programming (QP) and Disciplined Convex Programming (DCP).

2.2 Karush-Kuhn-Tucker and Global Optimality Conditions

Because the portfolio construction problems considered henceforth are convex, their solutions are exact and can be calculated via DCP. Consider objective function $f(w)$:

(4)

1. Mention normalizing weights
2. Denote objective function as something f(w)?
3. Discuss value in making function quadratic

2.3 Considerations when making constraints

1. risk modeling
2. noise in predictions
3. transaction costs
4. don't wish to make constraints overly binding
5. maintain convexity (brief, explain in full later)

2.4 Introducing all constraints

1. Max/min weights and individual turnover
2. Net exposure
3. attribute weights (Risk modeling)
4. Total turnover
5. gross exposure

2.5 Elaborating on Total turnover

1. Discuss value in total turnover
2. Explain math behind slack variables
3. assert convexity

2.6 Elaborating on gross exposure

1. Only talk about long-only, short-only (simple form)
2. mention kkt
3. identify convexity issue (definition of nonconvexity, try to make general)
4. solver characteristics
5. Alternate strategy
6. explain newfound convexity

3 Additional gating

3.1 value in penalties

1. Explain downsides of strict constraints
2. Introduce idea of gating

3.2 Alpha gating for total upper turnover

3.3 Elastic net

1. Show reason for needing elastic net and intuition
2. explain how the elastic net works
3. demonstrate convexity
4. Explain how exogenous parameters impact results and objective function
5. Briefly explain what other fields utilize elastic nets

3.4 Berhu

1. concept of huber's loss and what berhu's loss is
2. value of using berhu's loss
3. Converting Berhu loss into convex form and implementation
4. Explain how exogenous parameters impact results and objective function
5. Difference between Berhu loss and Elastic Net

4 Numerical Experiments

4.1 contents tbd

5 using huber's loss as an objective function

5.1 introduce huber's loss as objective function and functions

6 Conclusion

6.1 Parallels with Machine Learning

6.2 Next steps in research

References

[1] Author, A., & Coauthor, B. (2026). Title of foundational scientific paper. *Journal of Computational Science*, 12(3), 101–115.

[2] Developer, C. (2025). *Convex Optimization Solvers and Algorithms*. Academic Press.